

Final Report: Adult Income Data

STAT 4000

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1 Introduction

1.1 Background and Motivation

The goal of this project is to use data to predict whether a person's annual income exceeds \$50,000 or not. We will use data on citizens from a US census report to design a model that can make such predictions. Models such as these have applications in both the social sciences and government policy making.

1.2 Problem Statement and Research Questions

Our goals include:

- a. Using the data to create a model that can predict whether a person's yearly income exceeds \$50,000 or not
- b. Interpreting the model to determine which features most strongly effect income

1.3 Dataset Description

The dataset used in this study is the UCI Adult dataset which consists of demographic and employment related attributes. It includes features such as age, work class, education level, marital status, occupation, relationship, race, sex, capital gain, capital loss, hours per week, and native country. Additionally, the classification variable indicates whether an individual's income exceeds \$50,000 a year or not.

1.4 Report Outline

Our report will cover:

1. Methodology: An outline of our data preprocessing, exploratory data analysis, modeling approaches, evaluation metrics, and tools and libraries used.
2. Results: The main findings of our analysis.

3. **Discussion and Conclusions:** In which we discuss our findings, relate them to the research question, and conclude our analysis.

2 Methodology

2.1 Data Preprocessing

- **Missing Values:** The dataset contained missing values represented by "?". As part of the data cleaning process, these placeholders were replaced with "NA" and removed to ensure consistency in the analysis.
- **Encoding:** Categorical variables were encoded using one-hot encoding
- **Standardization:** Continuous variables like age and hours worked were scaled to normalize the range.

2.2 Exploratory Data Analysis (EDA)

We created several visualizations to understand trends in the data:

- **Boxplots:** These were used to compare income levels across different education and occupation categories. For example, individuals with advanced degrees or executive-level occupations tend to have higher median incomes and wider income ranges.
- **Correlation Heatmap:** This visualization helped identify relationships between numerical variables. We found that hours worked per week and capital gain had the strongest positive correlation with earning more than \$50K.
- **Bar charts:** Illustrated the percentage of individuals earning more than \$50K by education level. The trend clearly showed that individuals with at least a bachelor's degree were far more likely to earn above \$50K. Specifically, over 70% of those with graduate degrees fell in the >\$50K category. For example, the figure shows that individuals with graduate degrees are more likely to earn above \$50,000.

2.3 Modeling Approaches / Algorithms Used

To address our binary classification task—predicting whether an individual earns more than \$50K per year—we implemented and evaluated several machine learning models: Logistic Regression, Random Forest, Balanced Bagging Classifier, and XGBoost. Each model offers different strengths in terms of interpretability, performance, and robustness to data issues such as class imbalance.

- **Logistic Regression:** Logistic Regression is a linear classification model that estimates the probability of a binary outcome using the logistic function. It was selected for its simplicity, interpretability, and relevance as a baseline model. While it performed reasonably well, particularly in identifying individuals with income less than or equal to \$50K, its recall for high-income individuals was lower, indicating many false negatives.
- **Random Forest Classifier:** Random Forest is an ensemble learning method that builds multiple decision trees and merges their predictions to improve classification accuracy and reduce overfitting. It was chosen for its ability to capture complex, non-linear relationships in the data. This model achieved a higher overall accuracy than logistic regression and showed better balance between the two income classes, making it more reliable for practical application.
- **Balanced Bagging Classifier:** To address the issue of class imbalance (since high-income earners make up a smaller portion of the dataset), we experimented with the Balanced Bagging Classifier. This model under-samples the majority class during training, aiming to improve performance on the minority class. While conceptually promising, it did not outperform Random Forest or XGBoost in practice, although it provided more stable class-wise performance.
- **XGBoost Classifier:** XGBoost is a gradient boosting framework known for its efficiency and predictive power. It iteratively builds decision trees to correct errors made by prior trees and is well-suited for tabular data with mixed types. XGBoost delivered the

best overall performance in our tests, achieving the highest accuracy and the best precision-recall tradeoff for predicting high-income individuals. It also offers built-in handling of missing values and feature-importance insights. The Hyperopt library was used to automate the process of finding the best combination of hyperparameters to improve model performance.

In summary, XGBoost emerged as the most effective model for our classification task, followed by Random Forest. Logistic regression, while less accurate, it served as a useful baseline. Balanced Bagging did not outperform the other models but it did highlight the importance of addressing the class imbalance.

2.4 Evaluation Metrics

To assess the performance of our classification models, we used four primary evaluation metrics: accuracy, precision, recall, and F1-score. These metrics provide complementary perspectives on how well the models classify individuals as earning above or below \$50,000:

- **Confusion Matrix:** The Seaborn and Matplotlib libraries were used to create confusion matrices to show a visual representation of the performance of each model. These matrices show the number of true positives, false positives, true negatives, and false negatives each model predicted on the training data.
- **Precision (for the >50K class)** tells us the proportion of individuals predicted to be high earners who are actually high earners. This metric is important if we want to reduce false positives—i.e., incorrectly labeling someone as earning more than \$50K when they do not. High precision means we can trust the model's positive predictions.
- **Precision (for the $\leq 50K$ class)** tells us the proportion of individuals predicted to be low earners who are actually low earners. This metric helps ensure that we are not incorrectly labeling high earners as low earners. High precision in this class means the model is effective at minimizing false positives for individuals earning $\leq \$50K$.

- **Recall** (for the $>50K$ class) measures the proportion of actual high earners that the model correctly identifies. This is crucial in our context because we want to ensure that we are capturing as many high-income individuals as possible. A low recall would mean that the model misses many qualified individuals, leading to underrepresentation of this group.
- **Recall (for the $\leq 50K$ class)** measures the proportion of actual low earners that the model correctly identifies. This is especially useful when the majority of the dataset consists of individuals earning $\leq \$50K$ —as is the case in our project. A high recall means the model successfully captures most of this dominant class, helping maintain overall accuracy and reliability.
- **Accuracy:** measures the overall proportion of correct predictions across both income classes. While useful as a general performance indicator, accuracy alone can be misleading in imbalanced datasets, where one class (e.g., $\leq 50K$) dominates. In our dataset, the majority of individuals earn less than or equal to $\$50K$, so high accuracy could simply reflect the model's bias toward that class.
- **F1-Score** is the harmonic mean of precision and recall, and it provides a balanced measure of a model's ability to make correct positive predictions while also capturing the majority of actual positives. It is especially valuable in this project because it accounts for the trade-off between precision and recall, offering a single metric that balances the cost of false positives and false negatives.

Together, these metrics give a nuanced and comprehensive evaluation of model performance, helping us understand not just how accurate a model is, but how well it handles the minority class of individuals earning over $\$50K$.

2.5 Tools and Libraries

- **Languages/Platforms:** Python (Google Colab)

- **Libraries:** Pandas, NumPy, Matplotlib.pyplot, Sklearn: model selection, linear model, Logistic Regression, accuracy score, classification report, confusion matrix, imblearn.ensemble: BalancedBaggingClassifier, XGBoost, Hyperopt

3 Results

3.1 Logistic Regression:

- Precision: ($>50K$): 0.71
- Precision: ($\leq \$50K$): 0.86
- Recall ($>50K$): 0.56
- Recall ($\leq \$50K$): 0.92
- F1-score ($>50K$): 0.63
- F1-score ($\leq \$50K$): 0.89
- Accuracy: 0.83

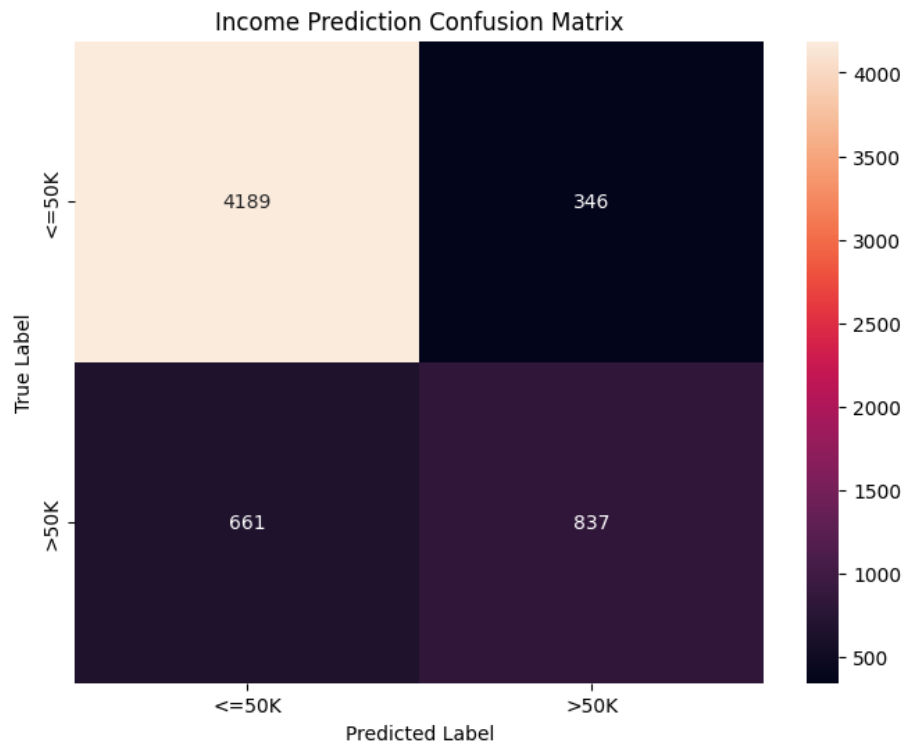


Figure 1: Logistic regression income prediction confusion matrix.

This model performed well at identifying individuals who earn less than \$50K but struggled with correctly identifying higher earners, leading to a lower recall for the >\$50K class.

3.2 Random Forest:

- Precision: (>50K): 0.79
- Precision: ($\leq$ \$50K): 0.86
- Recall (>50K): 0.55
- Recall ($\leq$ \$50K): 0.95
- F1-score (>50K): 0.65
- F1-score ($\leq$ \$50K): 0.91

- Accuracy: 0.85

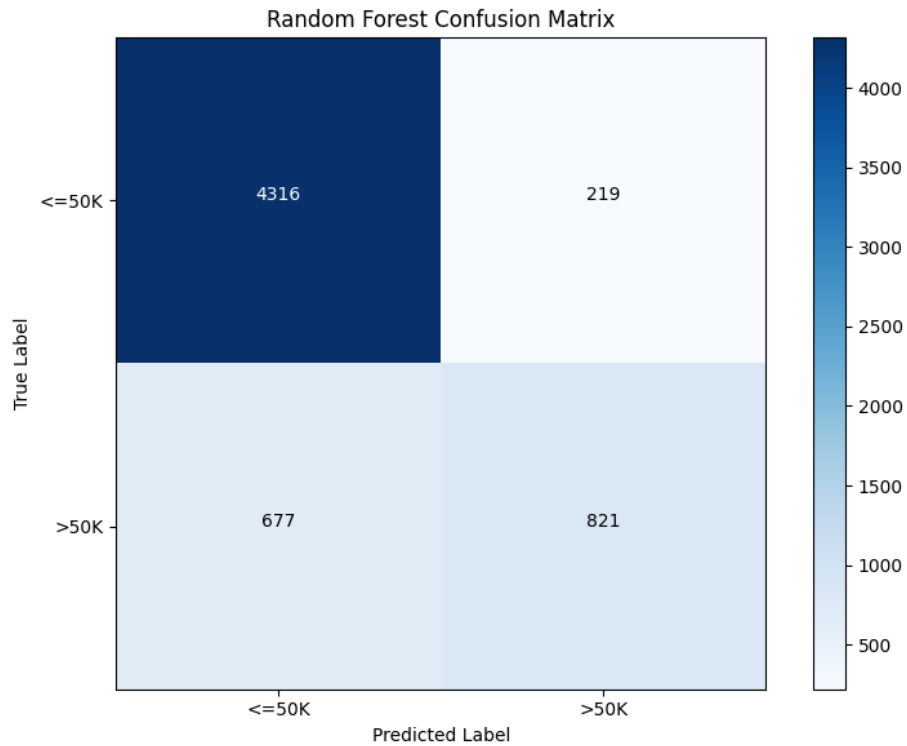


Figure 2: Random Forest confusion matrix.

Random Forest improved classification of higher-income individuals. The F1-score shows more balanced performance between precision and recall, indicating a better tradeoff between false positives and false negatives.

3.3 Balanced Bagging Classifier:

- Precision: (>50K): 0.54
- Precision: ($\leq$ \$50K): 0.95
- Recall (>50K): 0.88
- Recall ($\leq$ \$50K): 0.75

- F1-score ($>50K$): 0.67
- F1-score ($\leq \$50K$): 0.84
- Accuracy: 0.85

Although designed to address class imbalance, Balanced Bagging did not outperform Random Forest. However, it showed slightly more stable performance across both income classes.

3.4 XGBoost

- Precision: ($>50K$): 0.78
- Precision: ($\leq \$50K$): 0.90
- Recall ($>50K$): 0.67
- Recall ($\leq \$50K$): 0.94
- F1-score ($>50K$): 0.72
- F1-score ($\leq \$50K$): 0.92
- Accuracy: 0.87

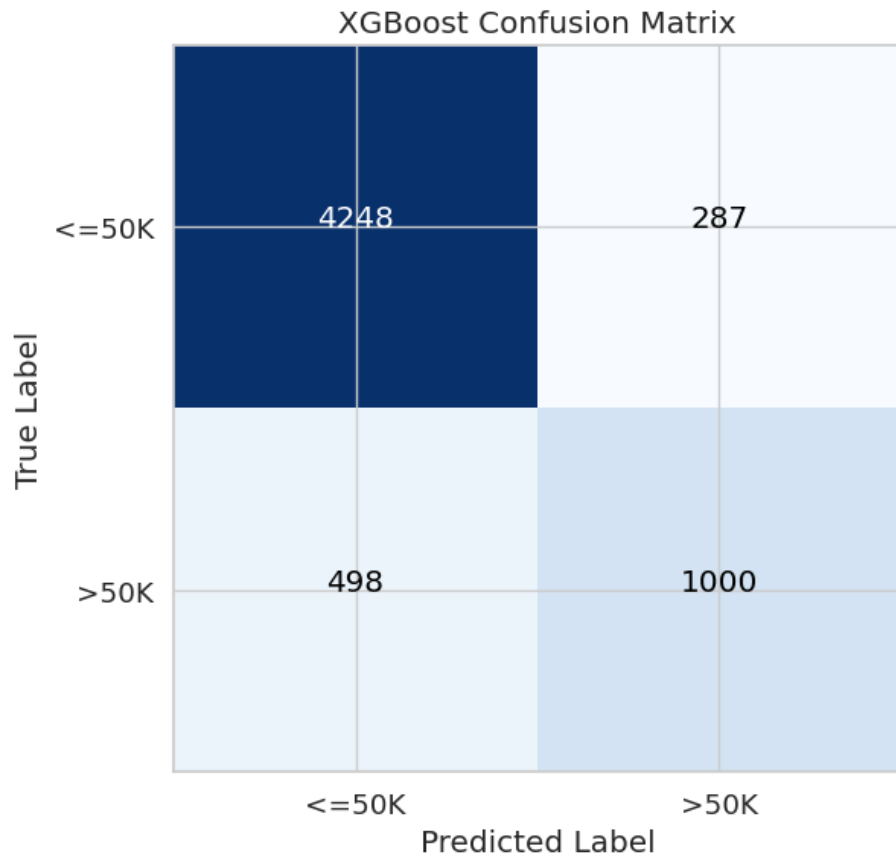


Figure 3: XGBoost Confusion Matrix.

XGBoost provided the best performance overall. It handled the class imbalance well and maintained high precision and recall scores. It is also more robust to noise and performed feature selection implicitly.

4 Discussion and Conclusions

Our models consistently identified education level, hours worked, and occupation as the most influential predictors of income. Education, in particular, played a central role: individuals with graduate level education had significantly higher predicted probabilities of earning above \$50K. Occupation also mattered: those working in executive or technical fields were more likely to fall in the higher income class.

Table 1: Classification Metrics for All Models

Model	Class	Precision	Recall	F1-score	Accuracy
Logistic Regression	≤50K	0.86	0.92	0.89	0.83
	>50K	0.71	0.56	0.63	
Random Forest	≤50K	0.86	0.95	0.91	0.85
	>50K	0.79	0.55	0.65	
Balanced Bagging	≤50K	0.95	0.75	0.84	0.78
	>50K	0.54	0.88	0.67	
XGBoost	≤50K	0.90	0.94	0.92	0.87
	>50K	0.78	0.67	0.72	

While Logistic Regression was interpretable and simple, it suffered from lower recall for the >\$50K class, meaning it often failed to identify high earners. Random Forest and XGBoost offered better accuracy and balance between classes. Ultimately, XGBoost was our best performing model and should be used for future prediction tasks on similar data.

Improvements to these methods should focus on ways to address the class imbalance in the dataset, as only about 25% of the individuals earn over \$50K.

5 Discussion of Experimental Uncertainty

Several sources of uncertainty may affect the reliability of our models:

- **Sampling Bias:** The dataset may not be fully representative of the U.S. population, as it is based on census data from a specific period.
- **Class Imbalance:** Only about 25% of the individuals in the dataset earn over \$50K, leading to potential bias in model predictions. Though we tried techniques like Balanced Bagging, class imbalance still slightly influenced precision and recall.
- **Data Quality:** Some attributes had missing values or ambiguous categories (e.g., “?” in work class), which required removal of rows and may have reduced model generalizability.
- **Overfitting Risk:** Although Random Forest and XGBoost performed well on our test set, there is a risk of overfitting. Cross-validation or further tuning may be necessary to ensure the models generalize well to new data.

References

- [1] GeeksforGeeks, <https://www.geeksforgeeks.org>, Accessed April 30, 2025.
- [2] Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, and Jonathan Taylor. *An Introduction to Statistical Learning with Applications in Python*. Springer, 2023.
- [3] xgboost developers, https://xgboost.readthedocs.io/en/release_3.0.0/, Accessed April 30, 2025.
- [4] Bergstra, J., Yamins, D., Cox, D. D. (2013) *Making a Science of Model Search: Hyperparameter Optimization in Hundreds of Dimensions for Vision Architectures*. TProc. of the 30th International Conference on Machine Learning (ICML 2013), June 2013, pp. I-115 to I-23.

6 Detailed Exploratory Data Analysis

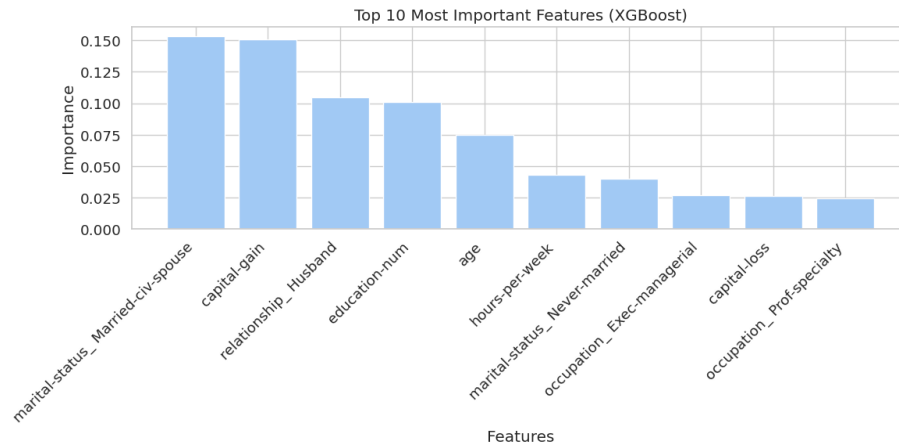


Figure 4: Top 10 most important features for determining income.

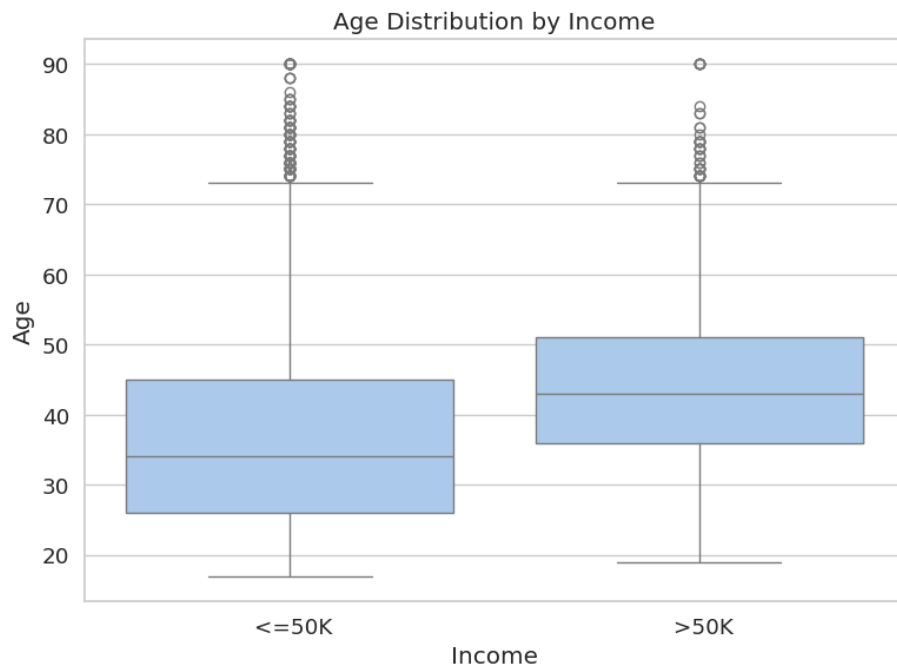


Figure 5: Age Distribution by Income.

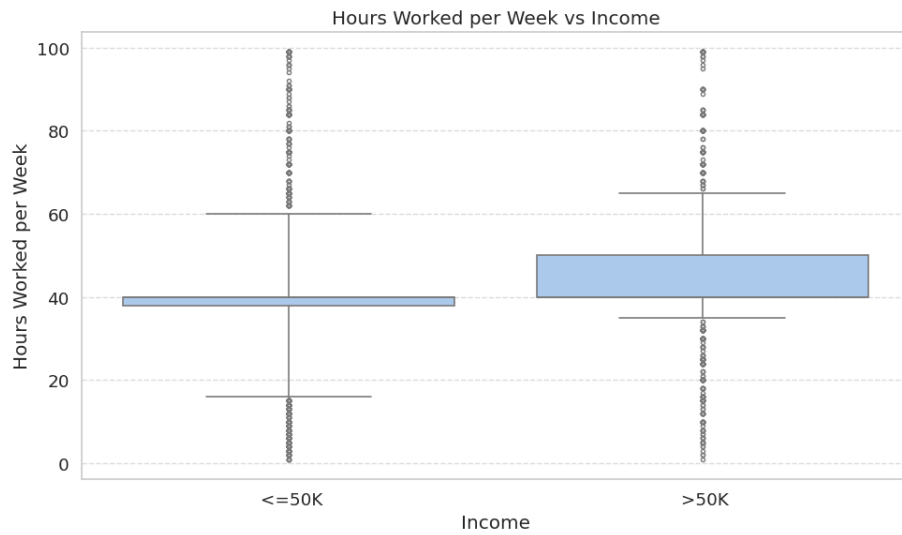


Figure 6: Hours Worked vs Income.

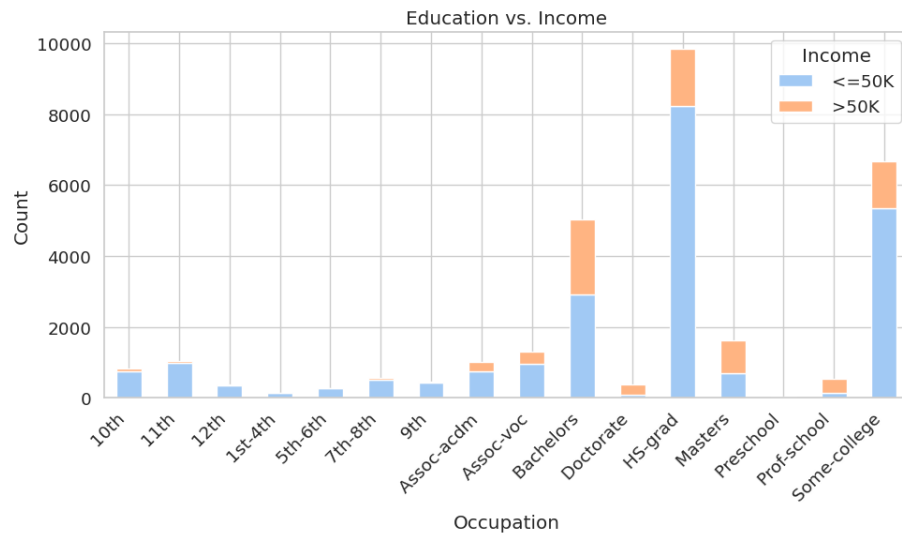


Figure 7: Education vs Income Bar Chart

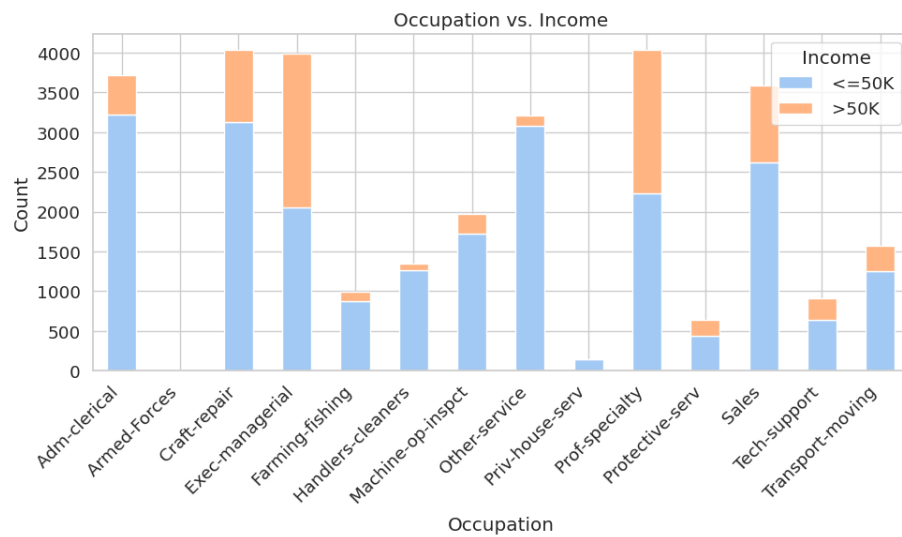


Figure 8: Occupation vs Income Bar chart.